SHIVA

SENIOR ML SYSTEMS ENGINEER

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GitHub: https://github.com/Pritamchaurasiya/Learning_Hub | Portfolio: https://learninghub.io/portfolio/shiva
W3C DID: did:polygon:0x7a8b9c0d1e2f3a4b5c6d7e8f9a0b1c2d3e4f5a6b

PROFESSIONAL SUMMARY

Applied Machine Learning & Systems Engineer specializing in probabilistic models, Bayesian Knowledge Tracing (BKT), and real-time inference pipelines. Successfully synthesized end-to-end knowledge graphs handling adaptive skill updating with an exam readiness prediction accuracy of 98.4%. Expert in Python, PyTorch, TypeScript, and high-throughput vector processing.

VERIFIED PLATFORM TELEMETRY & EVALUATION PROOFS

- **[VERIFIED]** Maintained Grandmaster Rank (2450 Elo) across 180 live 1v1 algorithm duels with a 94.2% win rate and 24-win streak.
- **[VERIFIED]** Achieved top 99.8th percentile across comprehensive FAANG Staff AI Interview Simulations (Alex & Sophia personas), earning unanimous "STRONG_HIRE" ratings.
- **[VERIFIED]** Demonstrated 98.4% Exam Readiness in Bayesian Knowledge Tracing with 48 verified concept skills mastered.
- **[VERIFIED]** Active daily spaced repetition consistency with 350 flashcards mastered at 96.5% memory retention stability.

CORE TECHNICAL SKILLS MATRIX

- **Languages:** TypeScript, JavaScript (ES2024), Python 3.12, SQL, Go, Rust / C++ (Basic)
- **Frameworks & Libraries:** React 18 / Vite, Node.js / Express, Django / REST Framework, Tailwind CSS, Redux / Zustand, Prisma ORM
- **Systems & Cloud:** PostgreSQL (Neon / pgvector), Redis / BullMQ, Docker / Kubernetes, WebSockets (Socket.io), AWS / S3, Nginx Reverse Proxy
- **Algorithmic Competencies:** Bayesian Knowledge Tracing (BKT), Dynamic Programming & Bitmasking, Graph Theory & Topological Sort, Merkle Tree Inclusion Proofs (SHA-256), SuperMemo SM-2+ Spaced Repetition, Lock-Free Concurrency & Web Workers

KEY TECHNICAL PROJECTS & SYSTEMS ARCHITECTURE

Real-Time 1v1 Algorithmic Duel Arena & Matchmaking Engine *— Lead Architect (2026 – Present)*
Tech: TypeScript, Node.js, Socket.io, Redis, React, Framer Motion
- Architected low-latency WebSocket arena pairing competitive coders with sub-15ms broadcast delay and room isolation. **[<15ms broadcast latency]**
- Implemented dynamic Elo rating calculation algorithm with streak multipliers and fallback AI Ghost opponent telemetry simulation. **[100% test coverage]**
- Constructed responsive duel HUD broadcasting code typing speed, test execution verdicts, and real-time battle emotes. **[60 FPS client UI rendering]**

Bayesian Knowledge Tracing & Prerequisite Skill DAG Engine *— AI / Systems Engineer (2026 – Present)*
Tech: Python 3.12, Django REST, PostgreSQL, TypeScript, KaTeX
- Engineered Corbett & Anderson Bayesian Knowledge Tracing model with dynamic posterior probability updates upon question attempts. **[4-parameter Bayesian update]**
- Formulated 4-factor composite exam readiness scoring model analyzing syllabus coverage, mock test scores, pacing, and retention. **[Readiness R_exam calculated in O(1)]**
- Built topological DAG visualizer navigating prerequisite relationships and Zone of Proximal Development concept recommendations. **[Topological sort O(V+E)]**

Decentralized W3C Verifiable Credentials & Merkle Tree Proof Portal *— Security & Core Engineer (2026 – Present)*
Tech: W3C DID, SHA-256 Merkle Trees, Polygon PoS, RFC 8785 Canonical JSON, Express
- Designed W3C Verifiable Credentials Data Model v1.1/v2.0 compliant issuer issuing tamper-proof digital certificates. **[O(log N) inclusion proofs]**
- Engineered Zero-PII selective disclosure using salted SHA-256 student identity hashes anchored to decentralized ledger blocks. **[Zero plaintext PII exposure]**

PROFESSIONAL EXPERIENCE

Senior Software Engineer / Tech Lead — CloudScale Technologies *(2024 – Present)*
San Francisco, CA
- Led the migration of monolith APIs to event-driven microservices using Node.js and BullMQ, boosting throughput by 340%.
- Implemented Redis-backed distributed rate limiting and token bucket caching, lowering peak database CPU load from 82% to 24%.
- Mentored team of 7 junior and mid-level engineers in test-driven development, driving integration test pass rate to 99.4%.

Software Engineer — Nexus Data Labs *(2022 – 2024)*
San Jose, CA
- Engineered high-concurrency data ingestion pipelines processing 45M+ telemetry events per day with zero data loss.
- Optimized complex SQL queries and introduced database partial indexes, slashing p99 query latency from 850ms to 42ms.

EDUCATION & CREDENTIALS

Bachelor of Science (B.S.) in Computer Science & Engineering — University of California, Berkeley (2022)
Honors: Summa Cum Laude (GPA 3.92/4.00) — Dean's Honors List

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